

Partial derivatives

One derivative per input — the only new skill is freezing variables.

THE IDEA

Every function so far had one input. Real functions have many: a loss function depends on every weight in the network. So instead of a single derivative, you get **one derivative per input**, and each is called a **partial derivative**.

Symbol	Means	How to compute it
$\partial f / \partial x$	How f changes when only x moves	Treat y as an ordinary number
$\partial f / \partial y$	How f changes when only y moves	Treat x as an ordinary number

The only new skill: freeze the other variable. The symbol ∂ is just "d" for partial — all the rules from Calculus (power rule, chain rule, product rule) apply completely unchanged.

TWO THINGS THAT TRIP PEOPLE UP

A term with **no x in it at all** is a constant when differentiating with respect to x , so it vanishes. And a term like x^2y is not a product needing the product rule — with y frozen, y is simply a multiplier sitting in front of x^2 .

WORKED EXAMPLE

PARTIAL DERIVATIVES — STEP BY STEP

a function with TWO inputs instead of one

THE IDEA

$f(x, y)$ depends on two things. so there are TWO derivatives :

$\partial f / \partial x$ = how f changes when only x moves (treat y as a NUMBER)

$\partial f / \partial y$ = how f changes when only y moves (treat x as a NUMBER)

the symbol ∂ is just "d" for partial. the rules are the same rules.

EXAMPLE $f(x, y) = x^2y + 3y^2$

STEP 1 — find $\partial f / \partial x$ (y is frozen, like a constant)

$$x^2y \rightarrow y \cdot 2x = 2xy$$

y is just a multiplier sitting in front of x^2

$$3y^2 \rightarrow 0$$

no x in it at all $\rightarrow$ it is a constant $\rightarrow$ vanishes

$$\partial f / \partial x = 2xy$$

STEP 2 — find $\partial f / \partial y$ (now x is frozen)

$$x^2y \rightarrow x^2 \cdot 1 = x^2$$

x^2 is the multiplier, y differentiates to 1

$$3y^2 \rightarrow 3 \cdot 2y = 6y$$

$$\partial f / \partial y = x^2 + 6y$$

THE ONLY NEW SKILL : freeze the other variable and treat it as an ordinary number. Everything else is chapter 3 and 4.

Why it matters for ML: a model with a million weights has a million partial derivatives. Each one answers the same question — "if I nudge this one weight, how much does the loss change?" — and that is exactly what backpropagation computes.

KEY POINTS FROM THIS TOPIC

Partial derivative	one derivative per input variable
$\partial f / \partial x$	freeze y , treat it as a number
$\partial f / \partial y$	freeze x , treat it as a number
Terms without that variable	become constants $\rightarrow$ differentiate to 0
Rules used	exactly the same as single-variable calculus

TEST — Topic 1

1. $f(x, y) = 3x + 4y$. Find $\partial f/\partial x$ and $\partial f/\partial y$. → **3 and 4**
2. $f(x, y) = x^2y^3$. Find both partial derivatives. → **$\partial f/\partial x = 2xy^3$ · $\partial f/\partial y = 3x^2y^2$**
3. $f(x, y) = x^2y + 5y$. Find both partial derivatives. → **$\partial f/\partial x = 2xy$ · $\partial f/\partial y = x^2 + 5$**